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# 1. Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
from sklearn.model_selection import train_test_split

# 2. Load Dataset
df = pd.read_csv("train.csv")
print("Dataset Shape:", df.shape)

# 3. Feature Selection and Data Preprocessing
# Convert categorical column 'Type' to numerical using One-Hot Encoding
df = pd.get_dummies(df, columns=['Type'], drop_first=True)

# Select independent features (X) and target (y)
features = ['Store', 'Size', 'Temperature', 'Fuel_Price', 'CPI', 'Unemployment', 'IsHoliday']
for col in df.columns:
    if 'Type_' in col:
        features.append(col)

X = df[features]
y = df['Weekly_Sales']

# Fill missing values with median
X = X.fillna(X.median())
y = y.fillna(y.median())

# Convert boolean column (IsHoliday) to integer (0 and 1)
X['IsHoliday'] = X['IsHoliday'].astype(int)

# 4. Split Data into Train and Test Sets (80% Training, 20% Testing)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 5. Train the Model
model = LinearRegression()
model.fit(X_train, y_train)

# 6. Make Predictions
y_pred = model.predict(X_test)

# 7. Print Model Performance Metrics
print("\n--- Model Performance ---")
print("Coefficients (Weights):", model.coef_)
print("Intercept:", model.intercept_)
print(f"R2 Score: {r2_score(y_test, y_pred):.4f}")
print(f"Mean Absolute Error (MAE): {mean_absolute_error(y_test, y_pred):.4f}")
print(f"Root Mean Squared Error (RMSE): {np.sqrt(mean_squared_error(y_test, y_pred)):.4f}")

# 8. Plot Graph: Actual vs Predicted Sales
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.3, color='blue')
plt.xlabel("Actual Weekly Sales")
plt.ylabel("Predicted Weekly Sales")
plt.title("Actual vs Predicted Weekly Sales")
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], color='red', linestyle='--', lw=2)
plt.show()
```

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Dataset Shape: (282451, 16)

--- Model Performance ---
Coefficients (Weights): [-1.32430704e+02  9.88168543e-02  1.49420870e+01 -8.49945550e+01
 -2.58895642e+01 -2.46507207e+02  1.28930769e+03 -4.09695644e+02
  5.73144299e+03]
Intercept: 10712.608183740402
R2 Score: 0.0694
Mean Absolute Error (MAE): 14397.8564
Root Mean Squared Error (RMSE): 21528.2455
```

